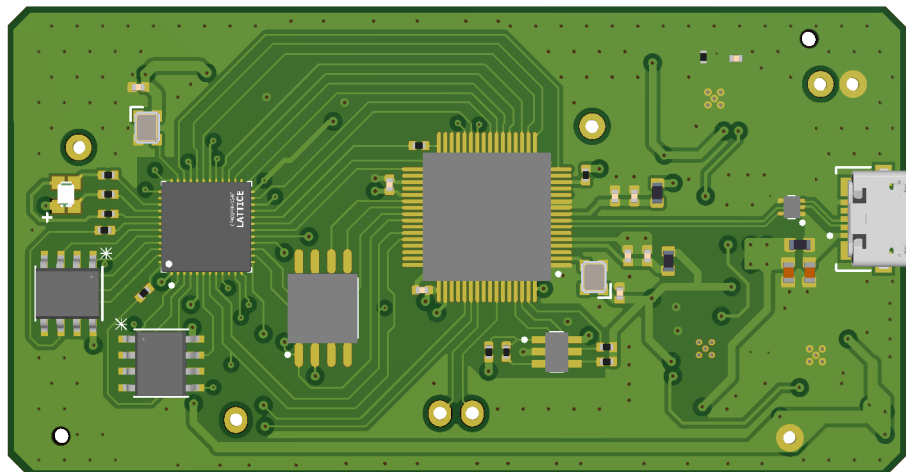


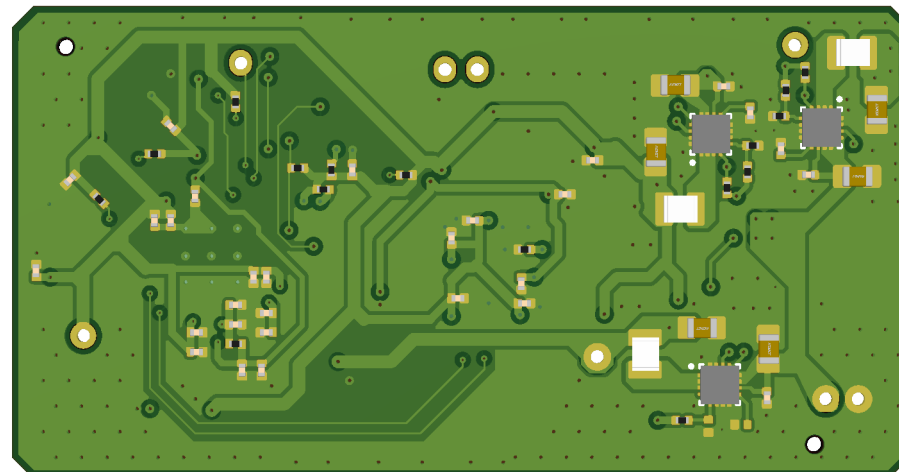
ICE40UP5K-SG48I Dev Board

Manufacturing Information

TOP



BOTTOM

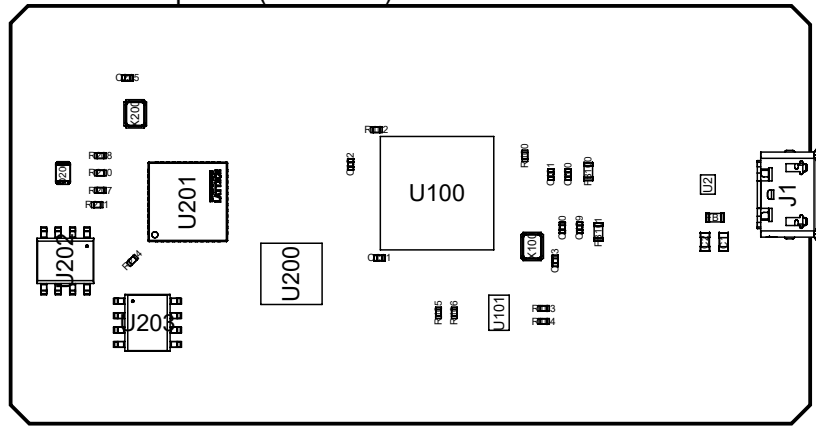


LENGTH = 70.485mm
WIDTH = 36.83mm
LAYERS = 4
THICKNESS = 1.6mm

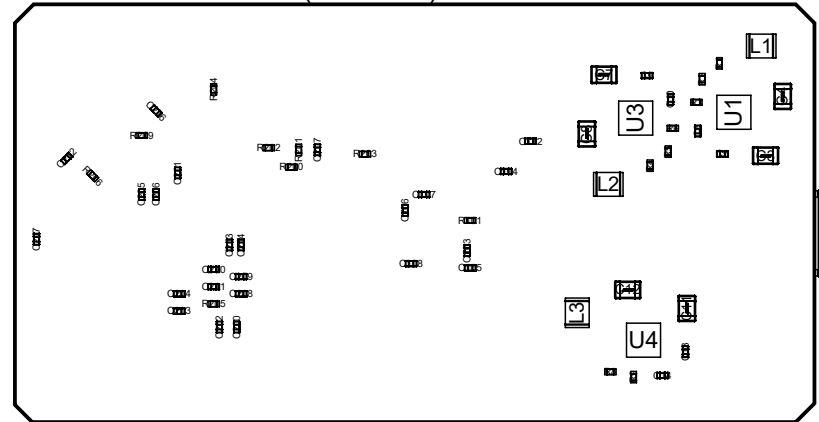
ICE40UP5K-SG48I Dev Board

ASSEMBLY VIEW

View from Top side (Scale 3:2)



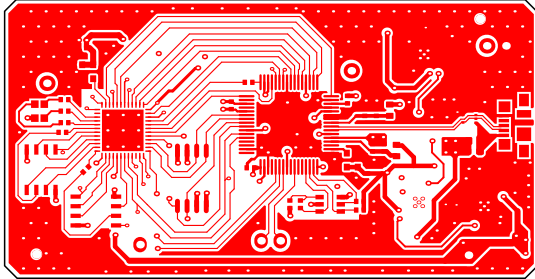
View from Bottom side (Scale 3:2)



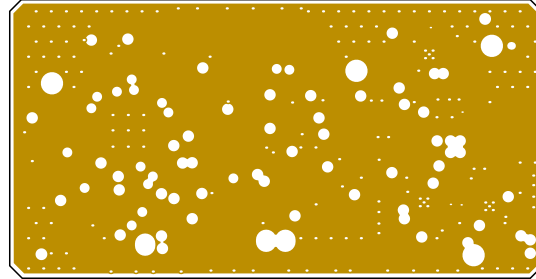
ICE40UP5K-SG48I Dev Board

COPPER LAYERS

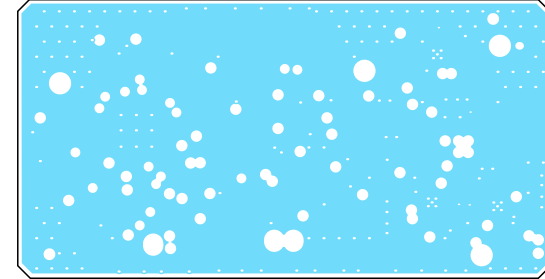
L1(SIG) (Scale 1:1)



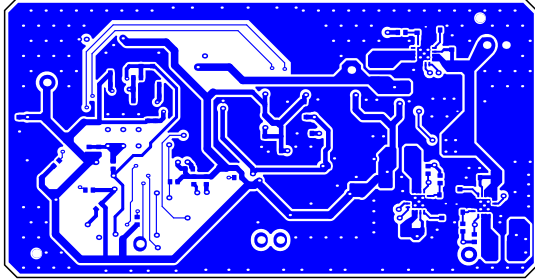
L2(GND) (Scale 1:1)



L3(GND) (Scale 1:1)



L4(SIG) (Scale 1:1)

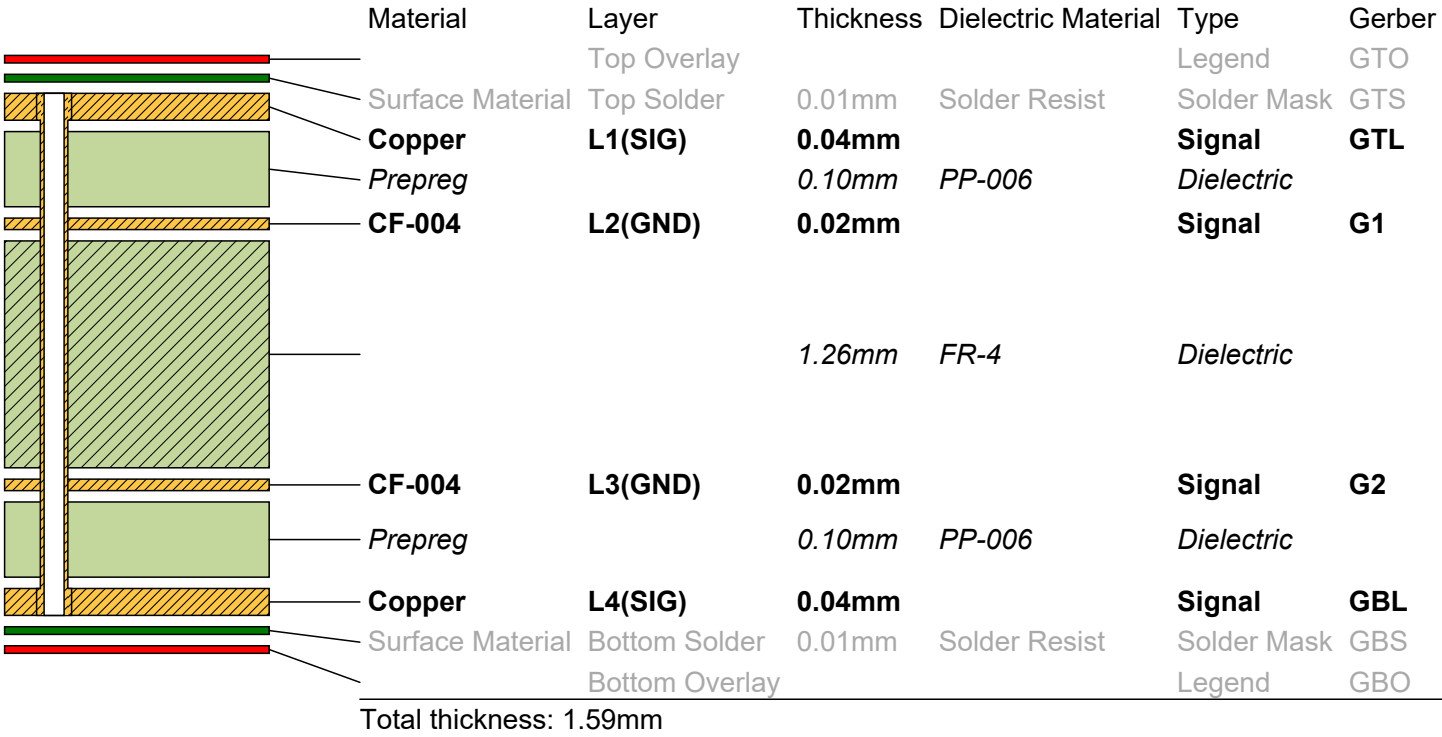


TOP = L1
BOTTOM = L4

ICE40UP5K-SG48I Dev Board

BUILD UP & IMPEDANCE

Layer Stack Legend



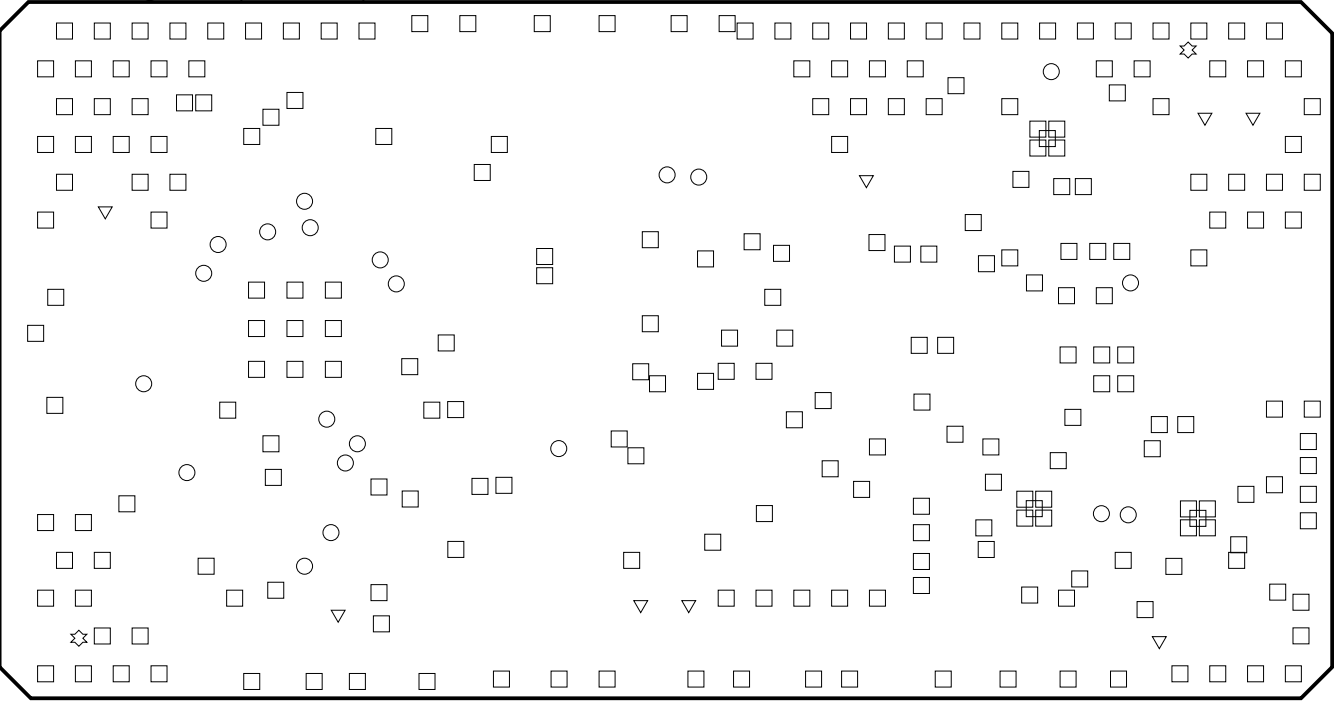
Transmission Line Structure Table

Impedance Id	Transmission Line	Target Impedance	Calculated Impedance	Trace layer	Wide Trace Width	Narrow Trace Width	Gap	Reference layers	Substack	Clearance	Target Tolerance
1	Coated Microstrip	50	52.16	L1(SIG)	0.16mm(6.16mil)	0.16mm(6.16mil)		L2(GND)	Board Layer Stack	0.13mm(5.00mil)	10%
2	Edge-Coupled Coated Microstrip	90	94.79	L1(SIG)	0.16mm(6.16mil)	0.16mm(6.16mil)	0.20mm(8.00mil)	L2(GND)	Board Layer Stack	0.13mm(5.00mil)	10%
3	Coated Microstrip	50	52.16	L4(SIG)	0.16mm(6.16mil)	0.16mm(6.16mil)		L3(GND)	Board Layer Stack	0.13mm(5.00mil)	10%
4	Edge-Coupled Coated Microstrip	90	94.79	L4(SIG)	0.16mm(6.16mil)	0.16mm(6.16mil)	0.20mm(8.00mil)	L3(GND)	Board Layer Stack	0.13mm(5.00mil)	10%

ICE40UP5K-SG48I Dev Board

DRILLS

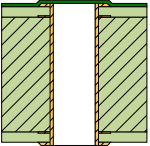
Drill Drawing View (Scale 5:2)



Drill Table

Symbol	Count	Hole Size	Plated	Hole Tolerance
○	21	0.20mm	Plated	
□	245	0.30mm	Plated	
▽	8	1.02mm	Plated	
☆	2	1.15mm	Non-Plated	
276 Total				

Via Type Type 1A (Scale 12:1)



Through-Vias only